

Research Onion Framework

The research onion framework is a structural model that helps researchers design their studies systematically and logically. This framework has the following layers.

Layer 01: Research Philosophy Layer 02: Research Approach Layer 03: Research Design Layer 04: Methodological Choices Layer 05: Time Horizon Layer 06: Techniques and Procedures	
--	--

Layer 01: Research Philosophy

Research Philosophy refers to the beliefs and assumptions about how **knowledge** (i.e. how the study answers the research questions) is created and what is considered valid knowledge.

Common research philosophies include:

- **Positivism** – Believes reality is objective and can be measured using numbers and facts.
- **Interpretivism** – Believes reality is subjective and understood through people's experience and meanings.
- **Pragmatism** – Focuses on what works best to answer the research problem and may combine different approaches.

Layer 02: Research Approach

The research approach examines how ***theory and data are linked*** in a study. Generally, there are three main research approaches. These research approaches are:

- **Deductive Approach** – Start with the existing **theory** and **test it using data**.
- **Inductive Approach** – Starts with **data** and builds new ideas or **theories** from observations.
- **Abductive Approach** – Moves back and forth between theory and data.

Layer 03: Research Design

Research design explains the overall plan of the study. The common research designs may include Survey research, case studies, experiments, and ethnography.

- **Survey research** – Collect data from many respondents using questionnaires.
- **Case Study** – Studying a single organisation, group, or event in detail.
- **Experiment** – Test cause and effect relationships under controlled conditions.
- **Ethnography** – Studies people in their natural environment.

Layer 04: Methodological choice

This layer explains what types of data the researcher will use. Generally, the following choices are available:

- **Quantitative Methods** – Use numerical data and statistical analysis.
- **Qualitative Methods** – Use non-numerical data such as words, opinions, and experience.
- **Mixed Methods** – Combine both quantitative and qualitative data.

Layer 05: Time Horizon

The time horizon refers to the **time frame** of the **study**. There are two main types:

- **Cross-sectional study**: Data is collected at **one point** in time.
- **Longitudinal study**: Data is collected over a **long period of time**.

Layer 06: Techniques and Procedures

This layer of the research onion outlines the methodological approach used to analyse the data. For example, to analyse the data, researchers can use **statistical**, **thematic**, or **content** analysis.